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 Studierichting: Informatica
 Oefeningenreeks 1D nr. 24.2

Opgave: Los de vergelijking $8z^6 + 7iz^3 + 1 = 0$ op in $\mathbb{C}$.

Oplossing:

Stel $Z = z^3$ dan is $8Z^2 + 7iZ + 1 = 0$.

$$\text{Hieruit volgt dat } Z_{1,2} = \frac{-7i \pm \sqrt{(7i)^2 - 4 \cdot 8}}{16} = \frac{-7i \pm 9i}{16} = \begin{cases} + : & \frac{i}{8} = \frac{1}{8} \operatorname{cis} \frac{\pi}{2} \\ - : & -i = \operatorname{cis} \frac{3\pi}{2} \end{cases}$$

$$Z_1 = z^3 = \frac{1}{8} \operatorname{cis} \frac{\pi}{2} \Rightarrow \begin{cases} z_1 = \frac{1}{2} \operatorname{cis} \frac{\pi}{6} = \frac{\sqrt{3} + i}{4} \\ z_2 = \frac{1}{2} \operatorname{cis} \frac{5\pi}{6} = \frac{i - \sqrt{3}}{4} \\ z_3 = \frac{1}{2} \operatorname{cis} \frac{9\pi}{6} = \frac{1}{2} \operatorname{cis} \frac{3\pi}{2} = -\frac{i}{2} \end{cases}$$

$$Z_2 = z^3 = \operatorname{cis} \frac{3\pi}{2} \Rightarrow \begin{cases} z_1 = \operatorname{cis} \frac{3\pi}{6} = \operatorname{cis} \frac{\pi}{2} = i \\ z_2 = \operatorname{cis} \frac{7\pi}{6} = \frac{-\sqrt{3} - i}{2} \\ z_3 = \operatorname{cis} \frac{11\pi}{6} = \frac{\sqrt{3} - i}{2} \end{cases}$$

Oplossing: De oplossingen voor $8z^6 + 7iz^3 + 1$ zijn $\left\{ \frac{\sqrt{3} + i}{4}, \frac{i - \sqrt{3}}{4}, -\frac{i}{2}, i, \frac{-\sqrt{3} - i}{2}, \frac{\sqrt{3} - i}{2} \right\}$

References